Assessment Grid — how you are graded

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Assessment Grid — Coding Agents Management

EPF · MDE S8 · September 2026

How this course is assessed

Band	What it means
Resit	The work was not done. This is an absence verdict, not a quality judgement.
Needs Work	The work exists but the competency is not demonstrated.
Basic	The competency is demonstrated mechanically. You did the thing; you could not yet do it under different conditions.
Solid	**The junior engineer I would be OK to hire.** You did the thing and you know why.
Outstanding	Judgement rather than compliance. You know where the method does *not* apply, and can say why.

The gap between **Basic** and **Solid** is *justification*. The gap between **Solid** and **Outstanding** is *knowing the limits*. Producing more artifacts moves you up the first gap, never the second.

What the grade is awarded on

There is one summative moment: a **6-minute individual oral** during Lab 3 (11 September), in front of the class, with the instructor acting as an interviewer.

The oral is not where your work is discovered. Your repositories are read in advance and the instructor arrives with a provisional band per competency. The interview **verifies** that the work is yours and that you understand it. A strong repository with an oral you cannot defend moves down. A modest repository defended with real understanding moves up.

Freeze: All work — C1, C2 and C3 — must be pushed by **20:00 on 10 September**. Anything pushed later is not read. There is one Freeze in this course and one date to remember.

The tooling question

You may use any coding agent — Claude Code, Cursor, Copilot, Codex, or another. The criteria below are written as **outcomes**, so they are satisfiable in any harness. Each competency also lists the **expected evidence**: the concrete artifacts most students will produce. Claude Code is taught as the worked example, so the expected evidence uses its vocabulary. If your harness does it differently, produce the equivalent and say so in the oral.

C1 — Frame and decompose work for a coding agent

> *Fiche statement: "Frame and plan agent tasks using an efficient combination of model and effort level."*

Turn a request that is not yet clear into work an agent can execute: sharpen it by interview, write it down as a specification, split it into units of work that can each be executed independently, and manage your context deliberately across the build.

A unit of work is a slice of the change that could be handed to a fresh agent session, with nothing else, and executed from its own description alone. That test is the definition. It has nothing to do with a unit test. The written description of a unit — an issue, a task file — is called its **ticket** throughout this course: one ticket per unit of work, and that is what is read when your repository is graded.

Subject: a change of your choosing in **your own repository**. It must be large enough to have needed decomposing — as a floor, **three or more units of work with at least one genuine blocking edge**. A change that decomposes trivially cannot demonstrate this competency, and picking one is itself a framing failure.

Expected evidence

- A specification document that is more than a restatement of the original request.
- Three or more tickets, one per unit of work, each executable on its own, with their blocking relationships written down.
- At least one recorded decision about context: where you started a fresh session, compacted, or delegated, and why. The "why" is what carries the mark — naming what you ruled out counts for more than naming what you did.

Band	Criteria
Resit	No specification and no decomposition. The work was a single prompt and whatever came back. You cannot describe how the work was scoped.
Needs Work	A document exists but restates the request without resolving anything. The work is not split into independently executable units, or the "units" are sequential steps of one indivisible task. Ordering is implicit. Context was managed by accident — you ran until quality degraded and then started over.
Basic	The work is split into three or more units with stated ordering, and the specification records what to build. At least one deliberate context decision exists, but it is justified after the fact rather than taken on purpose.
Solid	Each unit is genuinely independent: handed to a fresh agent session with no other context, it can be executed from its own description alone. At least one blocking edge reflects a **true dependency**, not merely a preferred order, and you can say what breaks if it is violated. The specification **resolves at least one ambiguity** the original request left open — you can name the question you had to answer to write it. Context boundaries were chosen deliberately and you can justify each one.
Outstanding	All of Solid, plus judgement about the method itself. You can name a place where you deliberately **did not** decompose, and why that was correct. Or the framing changed your mind: the interview revealed the original request was the wrong problem and you reframed it, with the evidence to show it. You can state what the method **costs**, not only what it buys.

Oral question: *"Walk me through how you framed this work. Show me one unit you would hand to a fresh agent session, and tell me what it blocks."*

C2 — Extend an agent's capabilities and constrain its behaviour

> *Fiche statement: "Extend agents capabilities and maintain control over them using skills and hooks."*

Two different kinds of control, and the competency is knowing which to use where:

- **Instruction** — a document the agent *reaches on its own* and follows. Persuasive, flexible, and probabilistic: the agent may not reach it, and may not follow it.
- **Enforcement** — a mechanism that fires *without the model's cooperation*. Rigid, narrow, and deterministic: it cannot be talked out of it.

Subject: your own repository, serving the C1 work.

Expected evidence

- **One instruction document** the agent reaches unprompted: a `SKILL.md`, a Cursor rule, an `AGENTS.md` / `CLAUDE.md` section. Not something you paste into a prompt.
- **One deterministic enforcement**: a hook, a pre-commit hook, a CI check.
- **Evidence that each actually triggered.** A transcript excerpt showing the agent reaching the document. A terminal capture showing the enforcement blocking something. This is non-negotiable at every band above Needs Work.

Band	Criteria
Resit	Neither artifact exists.
Needs Work	A document exists, but nothing gives the agent a reliable reason to reach it — it is a prose dump, or its trigger is vague. Or the "enforcement" is an instruction *asking* the agent to behave, which is not enforcement. No evidence that either fired.
Basic	Both artifacts exist. You can show the document being reached at least once, and the enforcement blocking something at least once.
Solid	The document's **trigger wording is deliberate**: you can say what makes it fire, and name a case where it should *not* fire. It contains a **checkable completion criterion** — the agent can tell done from not-done without asking you. The enforcement guards a genuine non-negotiable, and you can say **why that rule needs determinism** rather than instruction: what would happen if you had merely asked.
Outstanding	All of Solid, plus a limitation of your own work. You can name a case where your document fires when it should not, or fails to fire when it should — and either fixed it or explain the trade-off you accepted. Best evidence: you **observed a miss and tightened the wording**, and can show both versions.

Oral question: *"Show me the instruction you wrote. How do you know the agent reached it — and when would it not?"*

C3 — Recover from failure and feed the finding back

> *Fiche statement: "Apply repeatable methodology for diagnosing what went wrong and directing the agent toward a fix."*

The discipline is **the feedback loop before the hypothesis**. Reading code and theorising produces confident wrong answers. One command that goes red on *this* bug produces the right one.

Subject: the **shared repository**, handed to everyone at the same time, with the same bug, during the autonomy slot on 8 September. Everyone answers the same question, so answers are directly comparable. The slot opens the exercise; it does not close it — the Freeze does, at 20:00 on 10 September. You are asked to record whether it took you more than 1h30, which is not graded and is not a band criterion.

Expected evidence

- **One command** — a test invocation, a script, a CLI call — that you have run, that goes **red** on the bug and **green** when it is fixed.
- A stated cause, and the evidence that pointed at it.
- A regression test that fails on the original code.
- One sentence on where the finding belongs: the specification, an instruction document, or the code's structure.

Band	Criteria
Resit	No reproduction and no diagnosis.
Needs Work	You went straight to reading code and hypothesising. Any fix is unverified, or verified **only by the existing test suite** — which is exactly the trap in this repository. You cannot name a command that goes red on the bug.
Basic	You have one command that reproduces the bug and goes red on it. You found a cause. The fix is verified by that command. A regression test exists.
Solid	The loop is **tight**: fast, deterministic, and asserting on the **actual symptom** rather than "it didn't crash". You **rejected at least one plausible wrong cause** because the loop said so, and can name it. The regression test fails on the original code and sits at a seam that survives refactoring — it tests behaviour, not internals. You can say where the finding belongs and why.
Outstanding	All of Solid, plus you understand why the failure was *possible*. You can explain **why the existing test suite was insufficient** — what it asserted, and what it therefore could never catch. Or you identify the **class** of bug rather than the instance, and propose the enforcement that would have caught it before it shipped, connecting the finding back to C2.

Oral question: *"Show me the one command that went red on the sales pipeline bug. What did it rule out?"*

Resit

A **Resit** band means the work for that competency was not done. It is remedied by doing it: re-submit against the same grid, with a short written note on what changed. The other competencies' bands stand.

Academic honesty

The point of this course is that an agent does the work. Using one is not cheating; it is the subject. What is assessed is your management of it, and that is what the oral tests. You will be asked *why* — why this boundary, why this wording, why this loop. A student who directed the work can answer in a sentence. A student who did not, cannot, and no artifact compensates for that.

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